

DIGITAL PATHOLOGY

Presenter: Dr Ritusmita Baruah

PGT 1st year, Pathology

Moderator: Dr Ena Dowerah

Professor of Pathology, GMCH

OVERVIEW

- ▶ Digital pathology and its importance
- ▶ Components
- ▶ Systems and its key features
- ▶ Applications
- ▶ Advantages and disadvantages
- ▶ Future developments

What is Digital Pathology

- ▶ Digital pathology incorporates the acquisition, sharing and interpretation of pathology information including slides and data in a digital environment .
- ▶ Digital slides are created when glass slides are captured with a scanning device ,to provide a high resolution digital image that can be viewed on a computer screen or a mobile device .



- ▶ In digital pathology ,the microscope is replaced by a slide scanner linked to a reading station where the image data are assessed.
- ▶ Thus the pathologists no longer reads the physical slides under the microscope but the digital images on a monitor .



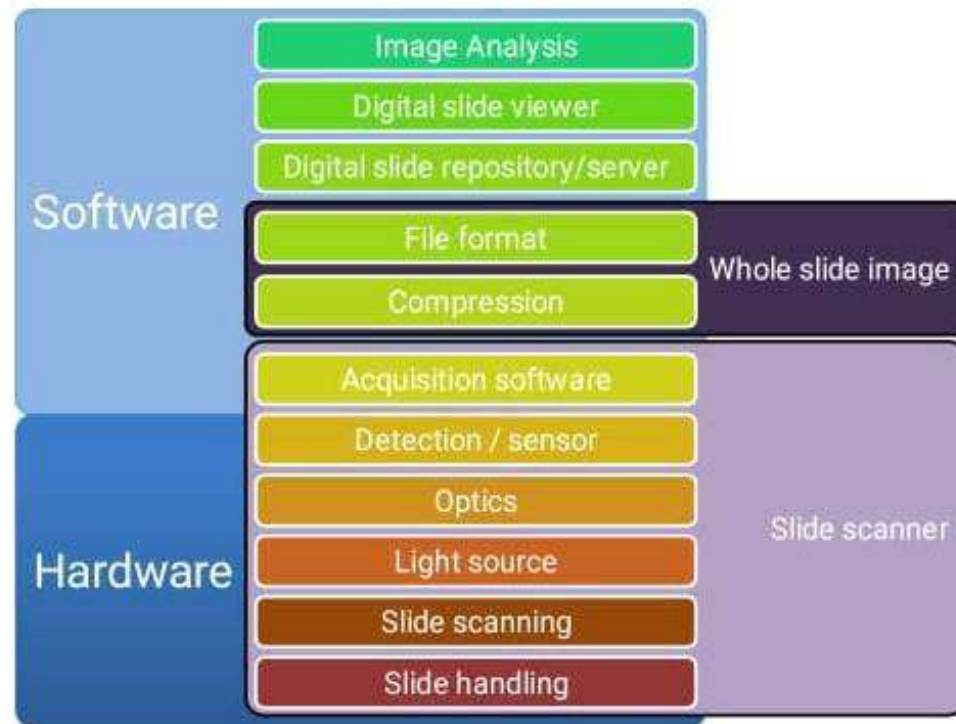
- ▶ Digital representation of slides enables histology images to be distributed, viewed and shared over computer network rather than relying on physical handling of glass slide .
- ▶ This is likely to represent the future of histology and cytology department.



WHOLE SLIDE IMAGING

- Refers to creation of a digital representation of the image presented by a glass slide at a level of detail comparable to that seen with a light microscope .

WSI: A layer model



DIGITAL IMAGE

- ▶ **Pixels** are the basic measuring unit in an image.They collectively create a digital image .
- ▶ In all commonly encountered digital image formats, pixels are squares although other shapes , in particular hexagons are in principle possible.



- ▶ **Resolution** is the term used to describe the crispness of an image .
- ▶ Higher the resolution, higher the pixel and sharper will be the image .
- ▶ **Pixel depth** refers to extent to which different colours can be distinguished.
- ▶ Although the data files which represent these images are large, software techniques enable these images to be viewed on a physical microscope.



When considering how images are represented 'perfect' fidelity may not be needed to convey meaningful information. The same original image is used throughout a to d. a) A normal full colour image as produced by most conventional digital cameras. b) The image modified to 256 shades of grey. c) The image represented as only 16 possible shades of grey with a much reduced file size, but some loss of image quality. d) With only 2 possible values per pixel, the image is still recognizable and able to convey meaning although it is significantly simplified and with a small fraction of the file size.

COMPONENTS

SLIDE SCANNER	WHOLE SLIDE IMAGING	OPINION AND ANALYSIS
SLIDE HANDLING	FILE FORMAT	DIGITAL SERVER
SCANNER	COMPRESSION	SLIDE VIEWER
OPTICS		IMAGE ANALYSIS
DETECTION		
SOFTWARE		

SLIDE SCANNER INCLUDE -

- ▶ Slide loading mechanism
- ▶ A motorized stage
- ▶ A light path
- ▶ Image capture device
- ▶ Software to create the composite image



CHOICE OF SCANNER

- ▶ Capacities from 1 to 400 slides
- ▶ Various slide handling mechanisms
- ▶ Software that drives the scanner
- ▶ 2 types of scanners
 - ▶ Microscope based / open
 - ▶ Box type scanner



MICROSCOPE BASED SCANNER

- ▶ Slower
- ▶ Have got algorithms

BOX BASED SCANNER

- ▶ Good at speed
- ▶ Doesn't have algorithms



Microscope based/open

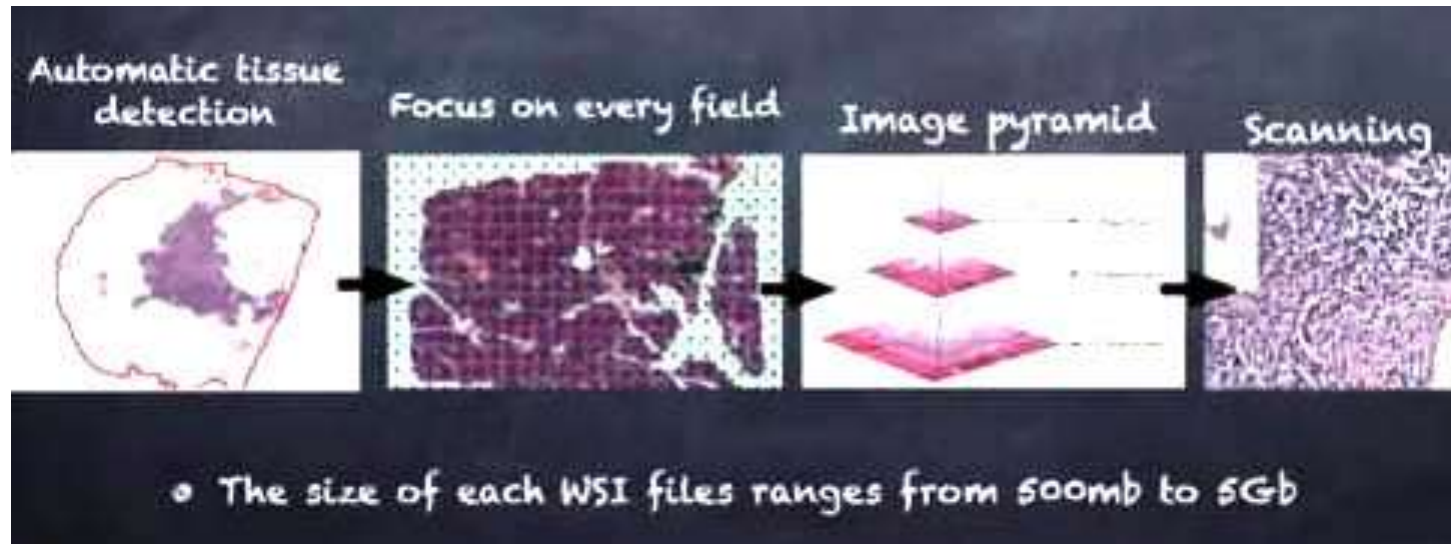


Box type scanner

BRIGHTFIELD SCANNER

- ▶ Most common type of scanner
- ▶ Least complicated, least expensive
- ▶ USES: H and E imaging, good for archive slides
- ▶ PROBLEMS: No Immunohistochemistry scoring, no algorithms for analysis





- ▶ The scanner include several parts designed to load glass slide ,a microscope,robotics to move the slides from the trays to the microscope and a barcode reader
- ▶ Stage moves continuously under a microscope ocular equipped with a high speed camera to capture each field .
- ▶ The slide moves in the X and Y planes ,focussing algorithms and robotics adjust the objective along the tissue Z plane topography .

PROCESS OF SCANNING

- ▶ Scanning by pyramid method is a unique feature .
- ▶ Zoom levels are precalculated and zoomed in the file .
- ▶ Prior to image acquisition the slide scanner may register the sample number by reading a barcode printed on the glass slide .
- ▶ **IMAGE OPTICS** - microscope objective and auxillary lenses ,optically transmit an image of the slide to the camera s sensor,- photosensitive elements (pixels) convert optical signals into digital signals .
- ▶ Scans the entire slide or region of interest
- ▶ Skip blank areas where tissue is absent to reduce scan time and file size .
- ▶ **SCAN TIME** - increases with tissue size, higher magnification and Z stacking .

IMAGE ACQUISITION

TILING

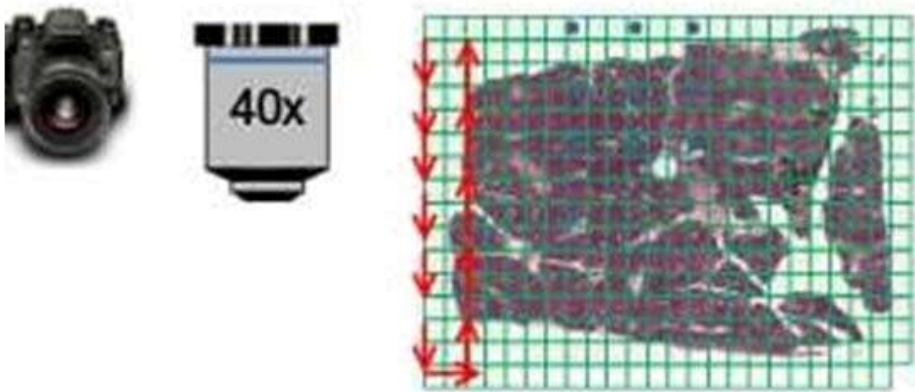
- ▶ The slide is scanned as a series of rectangular tiles .
- ▶ The tiles are assembled into a WSI either concurrently or after the scanning is finished.
- ▶ Autofocussing occur every nth field.
- ▶ Faster,simpler

LINE SCANNING

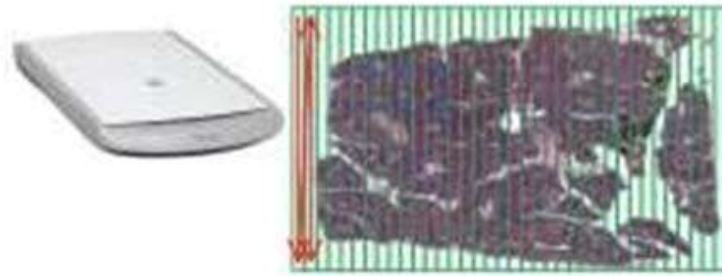
- ▶ The slide is scanned as a series of long,narrow strips.
- ▶ The strips are assembled into a WSI either concurrently or after the scanning is finished .
- ▶ Focus points are distributed only over the tissue forming a surface .
- ▶ Focus is then calculated for intervening tissue.

IMAGE ACQUISITION

Tiling



Line scanning



LIGHT SOURCE

- ▶ Traditionally halogen or xenon bulbs were used for WSI scanner
- ▶ Most vendors are moving to led light sources
- ▶ Considerations-
 - ▶ life time of bulb
 - ▶ fluctuation in output over time
 - ▶ implications for calibrations

OPTICS

- ▶ WSI scanner optics are similar but non identical to microscope optics
- ▶ Most scanners have a single microscope objective, typically a high quality 20x and 40x objective, combined with additional lenses to produce the final magnification .
- ▶ Scanners designed for live robotic telepathology usually have multiple objectives .
- ▶ Scanners with oil immersion objectives are available from a small number of vendors .
- ▶ Likewise scanners with very higher power objectives ie 60X, 100x are also available .

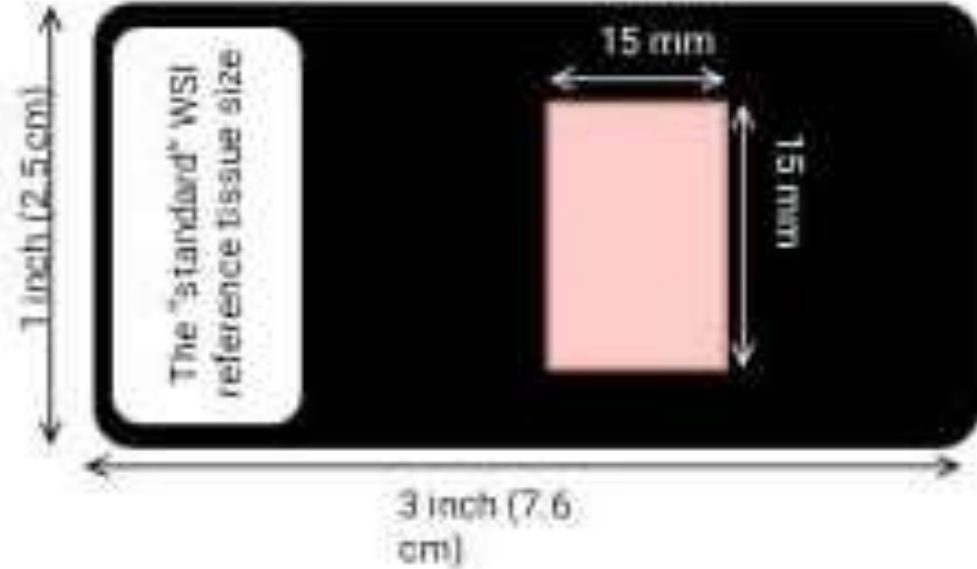
DETECTION/ SENSOR

- ▶ Brightfield scanners use CCD or CMOS chips .
- ▶ These chips have a grid of wells that accumulate photons to create an image .
- ▶ The wells correspond to pixels and can vary in size from chip to chip



SCAN SPEED

- ▶ Vendors tend to quote scan speeds in terms of a 15 mm x 15 mm piece of tissue and at a particular magnification equivalent eg 20x,40x
- ▶ Ranges from 45-270 secs .



WSI FILES

- ▶ Once all the image data acquired, they must be stored on discs.
- ▶ Hard disc drives offer high capacity and rapid access.
- ▶ Due to large size, they must be compressed .
- ▶ Compressed image formats such as JPEG or JPEG 2000 reduce the image file size using mathematical techniques to store the same data more efficiently.



FILE SIZE COMPRESSION

LOSS LESS COMPRESSION

- ▶ Exact original image obtained with no loss of details.
- ▶ Typically only able to reduce image file sizes around 2-3 folds.

LOSSY COMPRESSION

- ▶ Permanently discard information aiming to do so only for information which is non-contributory to the overall appreciation of the image .
- ▶ 50 fold higher compression ratio.

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- ▶ The UK's Royal College of Pathologists currently recommend that glass slides are held for 10 years .
 - ▶ Paraffin wax blocks must be held for 30 years .
 - ▶ Digital images used for primary diagnosis are retained for 10 years .
 - ▶ This is to ensure that the pathology remains available for subsequent review.

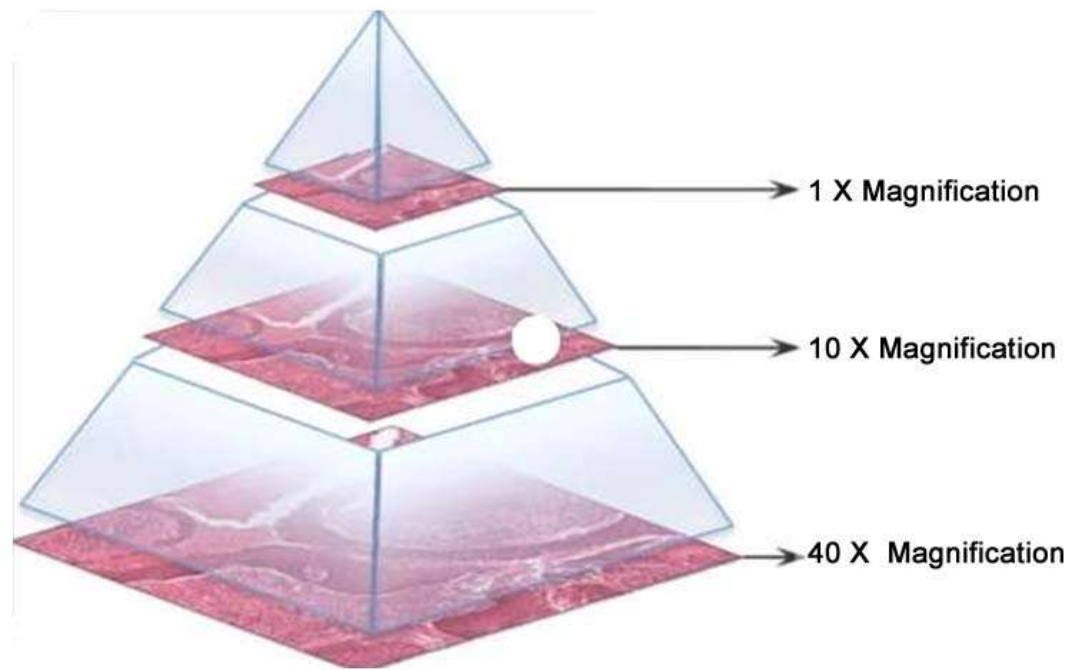
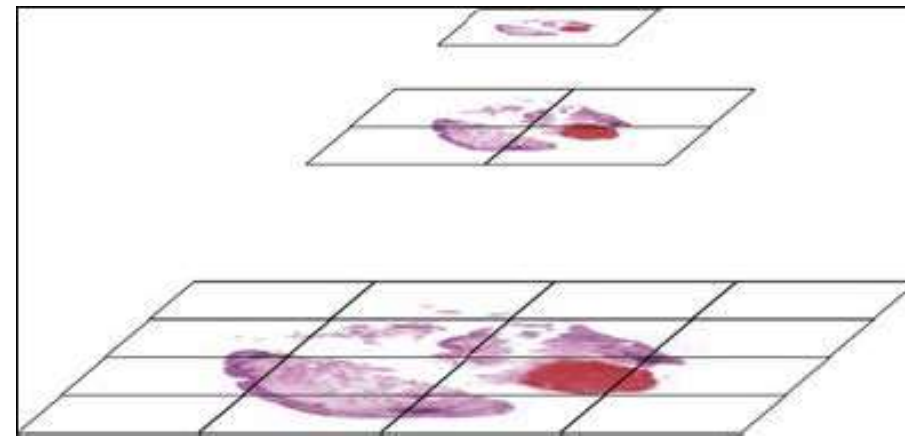


Fig. 22.2 Digital pathology file formats often use a pyramidal system. Multiple separate images are stored at different magnifications and resolutions. A pathologist viewing a low power overview image only needs to access the 'top' tier. As the magnification is increased, the relevant individual tiles from deeper tiers are displayed. This mechanism avoids the need to send entire image files across networks.



DIGITAL SLIDE REPOSITORY

- ▶ The Digital Slide Repository stores image data and metadata .
- ▶ It serves image data to a viewer client .
- ▶ Web client or thick client
- ▶ Many Digital Slides Repositories also support image analysis.

IMAGE VIEWER

- ▶ Digital slides are viewed on a monitor using image review software
- ▶ Installed directly on a user's workstation(THICK CLIENT) or can be accessed remotely via the web(THIN CLIENT)
- ▶ IMAGE STREAMING: When slides are being viewed ,software techniques are used so that only the part of the image being viewed at any one time is passed over the network,requiring only a fraction of the data transfer.
- ▶ Vendor softwares provide rich image viewing solutions
 - ▶ image navigation eg panning and zooming
 - ▶ image editing eg adding annotations, sharpening, colour manipulation.
 - ▶ basic image tools eg measurements ,snap shots,exporting images .

CLIENT SOFTWARE

- ▶ The software used by individual pathologists to request, view, navigate and manipulate WSIs.

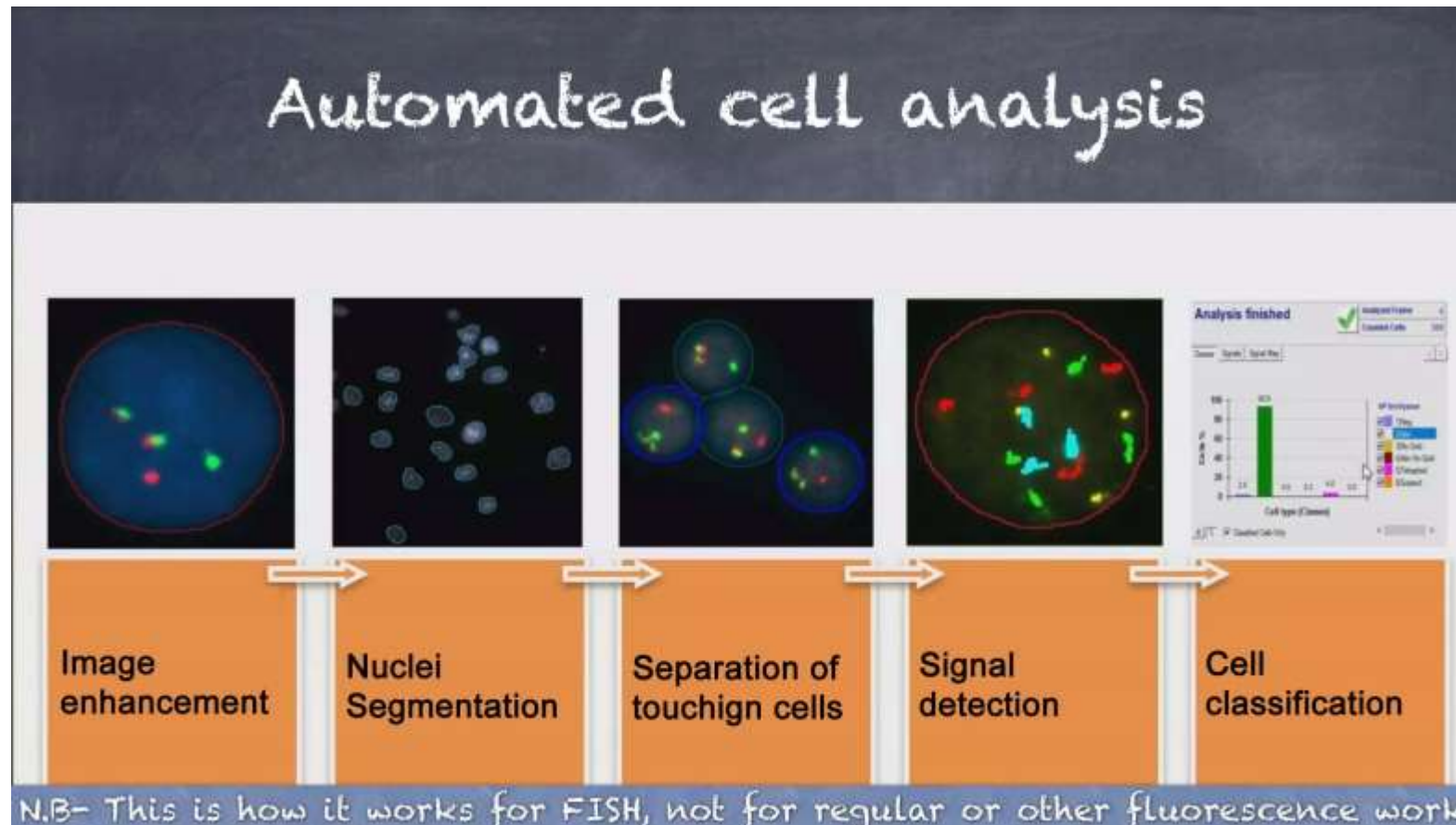
SERVER SOFTWARE

- ▶ Runs on the central computer responsible for streaming the stored images to clients on demand .

- Digital slides can be manipulated by a standard mouse but the greatest benefit experience is gained by use of 3D mouse that enables the pathologist to move around in three dimensions.

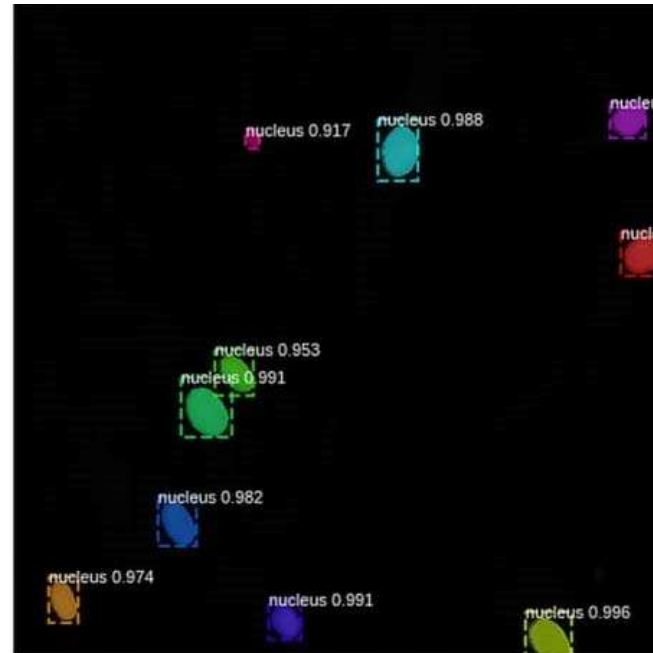


Analysis: AI based approaches for detection, segmentation, diagnosis and analysis of digitalized image.



NUCLEI SEGMENTATION

- ▶ It is a fundamental but challenging task in histopathology image analysis.
- ▶ Existence of overlapping regions which increases the difficulty of independent nuclei separation.
- ▶ Morphological changes in the cell nucleus are considered an important signal in many diseases ,specially in cancers and provide clinically meaningful information during diagnosis.



SPECIAL CASES

- ▶ **1) LARGE BLOCKS:** These mega or jumbo blocks can also be scanned, but the scanner needs to be designed to accommodate larger slides. There is an increase in slide acquisition time and file size .
- ▶ **2) CYTOLOGY PREPARATIONS:** The 3-dimensional nature of a typical cytology slide presents challenges. One cannot adjust the microscope to bring different depths of the preparation into focus on a digital image. This problem is addressed by acquiring multiple images of the cytology slide at different focal points which are treated as a stack of 2- dimensional images, a process called as Z - stacking where Z refers to the Z axis of a 3 dimensional image .
- ▶ **:3)FLUORESCENT SLIDES:** Acquiring digital images of slides stained with fluorescent stains will typically require additional hardware ,particularly a suitable light source and filters.Some vendors supply the fluorescence modules, or a dedicated scanner may be required.

APPLICATIONS

DIAGNOSTIC:

- ▶ 1) **INTRA OPERATIVE CRYO STAT (FROZEN SECTION) DIAGNOSIS** - one of the first applications where digital pathology has reached clinical practice.



This Hamamatsu scanner is used for capturing images for intraoperative frozen section diagnosis. The system holds just 6 slides but offers fast scan times which is appropriate for this application.

2) SECOND OPINION

- ▶ Ability to swiftly and easily seek a second opinion.
- ▶ The speed at which the digital image is viewable by a pathologist is an advantage and also savings on packing and postage .

3) MULTIDISCIPLINARY TEAM (MDT) MEETINGS AND CLINICO PATHOLOGICAL CONFERENCE .(CPC)

- ▶ Locating ,retrieving, refilling slides for regular MDT and CPC meetings,places a considerable burden on many labs .
- ▶ Digital images allow easy annotations of histological features of interest which can speed up case review and enable salient features to be easily displayed to the clinicians .

4) CLINICAL QUANTIFICATION

- ▶ Measurements of dysplasia, immunohistochemistry staining, percentage of tumour / stroma and glandular complexity are examples of features which a subjective observer, the pathologist must objectively ascribe.

5) TELEPATHOLOGY: It is a service performed by pathologist at a distance .

- ▶ It is the pathology service component of digital pathology.

NON - DIAGNOSTIC

- 1) Education and training
- 2) Research including image analysis
- 3) Quality assurance

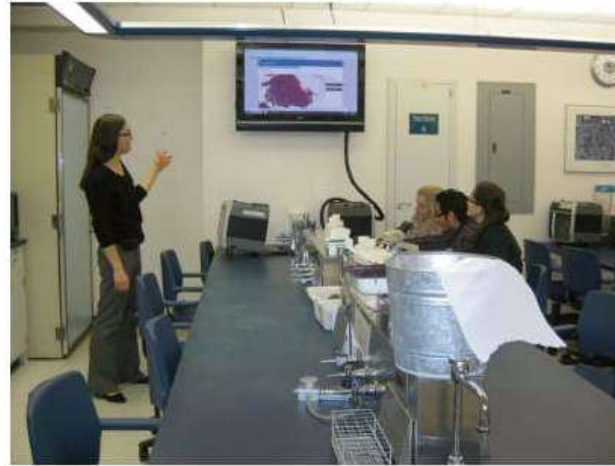


Fig. 7.1 The use of WSI to teach medical students enhances faculty and student interaction (image courtesy of Sara E. Monaco, MD from UPMC, USA)



Fig. 7.2 This pathology resident opted to study cytology using a digital slide teaching set instead of glass slides on a conventional microscope

EXTERNAL QUALITY ASSURANCE (EQA)

- ▶ Diagnostic EQA schemes involve the circulation of sets of slides to pathologists to ensure diagnostic uniformity. Where large numbers of pathologists are involved, multiple sections from the same tissue blocks may be required precluding the use of small biopsies or cases with only focal pathology.
- ▶ The use of digital slides avoid this problem allowing integration with web based systems for collating diagnosis.
- ▶ Digital slides are now widely used and accepted in diagnostic EQA schemes in the UK.
- ▶ The technology currently enabling eg approximately 500 pathologists to regularly participate in the UK Bowel Cancer Screening Programme scheme, with participants in the UK, Eire, Slovenia and Netherlands.

TO INCREASE EFFICIENCY

- ▶ Artificial intelligence: Digital pathology uses AI to automatically select tissue of interest /regions of interest (ROI).
- ▶ Stain normalisation: provision of templates to different labs.
- ▶ Pre trained system.
- ▶ Information fusion
- ▶ Experts from all over world to work collaborately .
- ▶ Production of high quality physical slides and quality control processes relating to the scanner.
- ▶ Installing scanner on a vibration proof table.
- ▶ Regular servicing and cleaning of slide scanner .



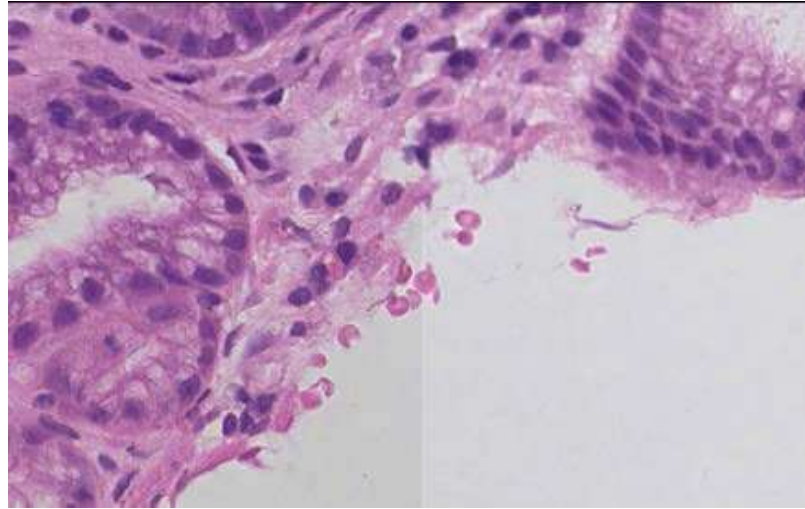
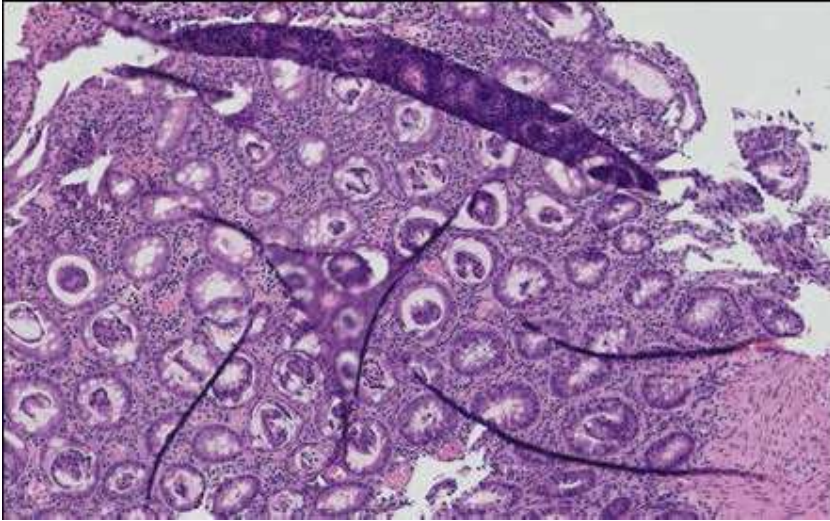
This high-capacity high speed scanner (Philips UFS) has been installed on a vibration-damping table to minimize artifact from a railway line running very close to the laboratory. (Courtesy of Tristan Brain.)

ADVANTAGES

- ▶ Networking with labs having the capacity and with top notch knowledge
- ▶ Sufficient work flow in less time.
- ▶ No transportation of blocks and glass slides.
- ▶ Consultations are easily made
- ▶ Clinico pathological conferences without glass slides.
- ▶ Measurements are easy using the built in ruler
- ▶ Automated Image analysis (prognostic,prediction)
- ▶ There is scope for the young pathologists.
- ▶ Work from home

DISADVANTAGES

- ▶ Expensive equipments
- ▶ Large storage component
- ▶ Lab without unstable technology, lack of it standards system unstable to support digitalisation
- ▶ A slide scanner reproduce all the artifacts including tissue folds, wax on the cover slip, air bubbles and tissue not covered by the coverslip. There are chances of mistakes, artefacts if not supervised.
- ▶ Blurring of image is an concern.



(a) Focusing artifact where the scanner has been unable to acquire a well-focused image of the entire tissue due to folds. Using a glass slide a pathologist may be able to compensate for this by focusing up and down through the depth of the tissue but such compensation is not possible on a digitally acquired image. **(b) Striping artifact caused by inconsistent illumination across the slide.** The scanner may require recalibrating.

DEVELOPMENTS

- 1) **BARCODING OF SLIDES:** the barcode must contain sufficient information to uniquely identify each slide.
- ▶ A variety of barcode format exists
 - ▶ One dimensional bar codes comprise a series of parallel black and white lines, while two dimensional formats e.g the quick response (QR) code is an array of black and white squares.



This slide bears a 2-dimensional barcode. This encodes data which uniquely identifies the slide and includes patient identifiers and details of the stain used. Use of such barcodes on slides is essential in high volume slide scanning applications to avoid time-consuming and error-prone manual image annotation.

2) REPORT CREATION :

- ▶ If pathologists are interacting directly with slide images on screen ,it may be logical for them to enter and sign out the final report concurrently where possible,perhaps using voice recognition software with a standardized canned text phrase or a reporting template. This is not established yet .

3)ADOPTION OF PAPERLESS REQUESTING AND REPORTING

- ▶ The benefits of digital distribution of slides are lost if paper request forms must be distributed to pathologists to enable them to report .
- ▶ Initial clinician's request can be made electronically or that paper request forms are scanned on receipt with the scanned request form image presented to the pathologists as slides are viewed.

4)STAFF TRAINING AND REPROFILING

- ▶ Requirement to load and unload slide scanners and there will be a need for equipment trouble shooting and maintenance with significant IT support .

5) PATHOLOGIST WORKSTATION

- ▶ Digital pathology workstations need to remain separate from the pathologist's standard PC used for email and other routine office tasks .

6) PATHOLOGIST TRAINING

- ▶ Training in the use of the software chosen will be required.
- ▶ Trainings should be documented and repeated at intervals .

7) APPLICATION OF SPECIFIC VALIDATION STUDIES

- ▶ **Validation** is required for each application of a digital slide.
- ▶ **Eg** general gastrointestinal pathology , neuropathology
- ▶ **Separate validation studies** may also be required for special stains, immune histochemistry ,frozen sections and cytology preparations .

8) MOBILE DEVICE

- ▶ Smart phones and other mobile devices offer a great potential to perform telepathology as they are relatively cheap compared to commercial slide scanner ,they are portable,provide internet connectivity, posses excellent digital cameras and can be easily attached to a microscope.

DIGITAL PATHOLOGY ASSOCIATION

- ▶ Does not limit its term to computer based methodologies for image analysis, but rather as a field of pathology that uses AI methods to combine pathologic images and metadata from a variety of related sources to analyse patient specimens.
- ▶ Non profit organisation of professional in the field of pathology and related technologies.

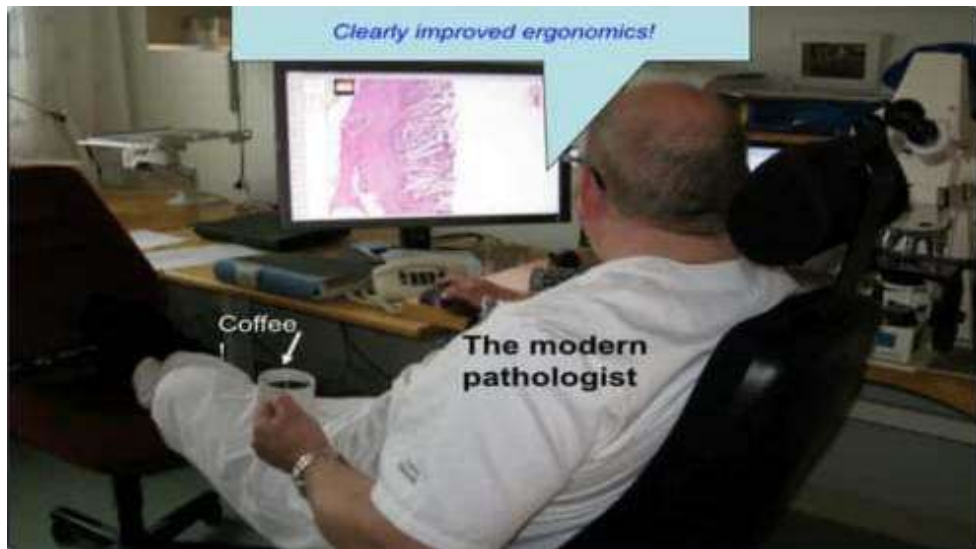


VALIDATION AND REGULATORY ISSUES

- ▶ Validation of a WSI system is essential prior to its use for routine diagnostic work.
- ▶ A validation study aims to demonstrate that a diagnosis from a digital slide has at least the same accuracy as that from a glass slide.
- ▶ The Digital pathology association and the college of American pathologists have both published guidelines on the conduct of validation studies.

SUMMARY

- ▶ Automation of tasks currently carried out by human brains will truly revolutionise cellular pathology in the path of cancer diagnosis and become pivotal in the new era of personalised medicine.
- ▶ At the same time we should not forget that the new possibilities will also create new dependencies.



THANK YOU

